

KENAI 29 ENE, AK, USA

WMO: 703420

Lat: 60.7237N

Lon: 150.4484W

Elev: 86

StdP: 100.30

Time Zone: -9.00 (NAK)

Period: 10-19

WBAN: 26563

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.1	-26.5	-33.0	0.2	-30.2	-29.1	0.3	-26.5	4.0	1.1	3.5	-0.1	0.0	N/A	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	24.5	15.9	22.5	14.8	20.5	13.8	16.6	23.1	15.4	21.1	14.5	19.2	1.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.7	9.9	18.0	13.0	9.4	16.7	12.2	9.0	15.9	46.5	23.1	43.2	21.3	40.8	19.3	19.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.8	3.1	2.6	-31.2	26.7	4.4	2.4	-34.3	28.5	-36.9	29.9	-39.4	31.3	-42.6	33.1		
(5)			-31.3	17.3	4.4	1.1	-34.4	18.1	-37.0	18.7	-39.4	19.3	-42.6	20.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-9.1	-6.8	-4.8	2.0	7.3	11.5	13.8	12.7	8.3	2.3	-6.5	-7.9
	DBStd	9.89	9.02	5.95	6.33	3.88	2.81	2.53	2.15	2.43	3.07	4.68	7.31	7.78
	HDD10.0	3219	592	470	458	240	89	11	1	4	64	239	494	556
	HDD18.3	5978	850	703	717	490	341	204	141	176	301	497	744	814
	CDD10.0	284	0	0	0	0	7	57	119	87	14	1	0	0
	CDD18.3	2	0	0	0	0	0	1	1	1	0	0	0	0
	CDH23.3	114	0	0	0	0	2	41	58	13	0	0	0	0
	CDH26.7	17	0	0	0	0	0	6	11	0	0	0	0	0
(14) Wind	WSAvg	0.9	0.6	0.9	0.9	1.1	1.2	1.1	1.0	1.0	0.9	0.9	0.7	0.7
(15) Precipitation	PrecAvg	901	79	68	49	42	42	47	66	98	128	103	87	92
	PrecMax	1090	145	180	90	97	104	93	115	179	237	219	175	142
	PrecMin	618	7	3	7	12	14	15	29	30	56	34	5	25
	PrecStd	98	37	33	22	18	23	18	19	37	49	41	50	31
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	7.1	9.9	13.3	22.8	27.4	28.7	24.9	19.3	13.0	8.1	7.1
		MCWB	4.2	3.9	5.1	6.7	12.5	16.3	18.5	16.7	12.4	8.6	4.5	3.8
	2%	DB	5.7	4.9	7.8	11.5	20.0	24.0	24.8	22.8	17.1	11.2	6.3	5.3
		MCWB	3.0	2.2	3.8	5.4	10.8	15.0	16.2	15.5	11.7	7.3	3.9	2.8
	5%	DB	4.2	3.2	6.0	10.3	17.1	21.4	22.8	20.2	15.5	10.1	4.3	3.6
		MCWB	2.1	1.2	2.7	4.7	9.4	13.4	15.6	14.0	11.0	6.7	3.0	1.8
	10%	DB	2.6	1.4	4.2	9.2	14.9	19.2	20.8	18.6	14.0	8.9	2.7	1.6
		MCWB	1.2	0.0	1.4	4.2	8.4	12.4	14.5	13.1	10.4	6.1	1.7	0.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	4.0	5.3	7.2	12.9	17.4	18.5	17.4	13.3	9.2	5.0	4.0
		MCDB	7.2	6.5	8.8	12.2	22.6	26.3	28.0	23.9	17.2	11.1	6.9	6.1
	2%	WB	3.1	2.4	4.0	5.9	11.3	15.3	16.9	15.8	12.4	8.3	4.0	2.8
		MCDB	5.2	4.4	7.4	10.4	18.9	23.1	23.6	20.9	15.6	10.2	5.9	4.8
	5%	WB	2.3	1.3	2.9	5.2	10.0	14.0	15.8	14.9	11.7	7.3	3.0	1.7
		MCDB	4.2	3.1	5.6	9.4	15.9	20.2	21.7	19.3	14.4	9.3	4.3	3.4
	10%	WB	1.1	0.2	1.7	4.5	8.9	12.7	15.0	14.0	11.0	6.5	1.7	0.4
		MCDB	2.6	1.4	4.0	8.7	14.2	18.2	20.1	17.4	13.4	8.5	2.7	1.4
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.6	12.3	11.3	12.4	12.1	11.2	10.4	10.1	9.0	8.6	7.8
		MCDBR	8.0	8.2	10.6	13.1	18.7	18.4	17.0	14.3	12.6	8.7	7.0	7.8
		MCWBR	5.7	6.1	6.9	7.7	10.2	9.7	9.2	7.8	7.4	5.6	5.1	5.6
	5% WB	MCDBR	7.6	7.8	10.0	11.5	15.9	16.0	15.0	12.0	9.9	7.5	6.6	7.4
		MCWBR	5.5	5.9	6.8	6.9	9.1	8.9	8.4	6.9	6.4	5.2	4.9	5.7
(40) Clear-Sky Solar Irradiance	taub		0.235	0.283	0.299	0.341	0.368	0.357	0.369	0.353	0.320	0.296	0.246	0.220
	taud		2.138	2.249	2.288	2.302	2.322	2.439	2.434	2.480	2.559	2.494	2.154	1.961
	Ebn at Noon		525	711	828	854	854	872	850	838	813	708	493	339
	Edh at Noon		39	66	88	106	112	102	100	89	69	52	37	28
(44) All-Sky Solar Radiation	RadAvg		0.29	0.96	2.54	4.11	5.17	5.40	4.75	3.77	2.45	1.16	0.42	0.15
	RadStd		0.04	0.14	0.30	0.40	0.46	0.38	0.47	0.45	0.35	0.13	0.07	0.02

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	+0.69	+2.09	+1.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page